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PRODUCT TEARDOWN DECK

The Broken Handshake: IRCTC Payment Failures

A product manager's analysis of the "Session Expired" friction point, covering user journey, heuristics, constraints, and solutions.

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Teardown Agenda

01. Context

User Persona, Journey Mapping & The "Aha!" Moment.

02. Analysis

Problem Identification, Heuristic Evaluation & System Constraints.

03. Solution

Strategic Trade-offs, Proposed Designs & Success Metrics.

User Persona

"The Anxious Tactician"

Name: Rohan (28, Software Engineer)

Goal: Secure 4 tickets home for Diwali. The window is only 15 minutes.

Pain Points:

- Fear of "Sold Out" status.
- Time pressure (Tatkal window).
- Repetitive data entry anxiety.

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User Journey Map

1	2	3	4	×	6
Login	**Search**	**Fill Form**	**Pay**	**Failure**	**Logout**
High Alert	Focused	Anxious	Hopeful	Panic	Frustration

The Drop-off Point: Between Step 5 and 6. The user expects to retry payment but is forced to Step 1 (Login).

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Problem Statement

The "Nuclear" Session Kill

When a payment gateway timeout occurs, the app treats it as a security breach rather than a process error.

Core Issue: Total loss of user input context. The user is penalized for a banking infrastructure failure.

Impact: 100% Data Loss.

Heuristic Evaluation

⚠ Violations

- **User Control & Freedom:** User has no "Emergency Exit" to go back to the payment selection. They are forced out.
- **Error Prevention:** System assumes malice instead of mistake.
- **Recognition rather than Recall:** User must recall/re-enter all passenger details.

✓ Adherence

- **Visibility of System Status:** The error message is clear ("Transaction Failed").
- **Consistency:** The logout behavior is consistent (albeit frustrating) across the platform.

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Root Cause Analysis (The 'Why')

Server Load

High concurrency. Every active session consumes RAM. Aggressive timeouts are a "Load Shedding" mechanism.

Inventory Locking

Seats are locked on "Pay". If payment fails, seats MUST be released instantly to prevent hoarding.

Security

Strict handshakes ensure no one manipulates the token. A broken link is treated as a threat.

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Competitive Landscape

Feature	IRCTC (Current)	MakeMyTrip (Aggregator)	Airline Apps (Indigo/AirAsia)
Payment Failure	**Session Termination**	**Retry Screen**	**Retry Screen**
Seat Inventory	Released Instantly	Soft Hold (5 mins)	Soft Hold (10-15 mins)
Data Persistence	**Wiped**	**Saved Locally**	**Saved in Session**
Security Model	Zero Trust	User Trust	User Trust

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Proposed Solutions

Solution 1: The "Grace Period"

Mechanism: Release inventory but keep session token alive for 120s.

UI: "Payment Failed. Seats released. Retry?"

High Impact

Solution 2: Client-Side Draft

Mechanism: Save passenger JSON in LocalStorage before payment.

UI: "Login Detected. Restore Draft?"

Low Effort

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Prioritization (RICE Framework)

Solution	Reach (Users)	Impact (0-3)	Confidence (%)	Effort (1-5)	Score
1. Grace Period (Server)	100% (All Failures)	3 (High)	90%	4 (Backend Change)	**67.5**
2. Client Draft (App)	80% (App Users)	2 (Med)	95%	2 (Frontend Change)	**76.0 (WINNER)**

Recommendation: *Start with Solution 2 (Client Draft) as an MVP due to lower effort and high score.*

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Success Metrics

Recovery Rate

+15%

Increase in users who successfully book after a first failure.

Time to Recover

-90s

Reduction in time taken to re-initiate booking.

Server Load

-10%

Reduction in Login API calls during peak hours.

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Summary

By decoupling "Inventory Locking" from "Data Persistence", IRCTC can maintain strict security while treating user effort with dignity.